Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Management Information System – (01CT0704)	**Aim:** Assignment - 1	
Assignment - 1	**Date:- 26-08-2025**	**Enrollment No:-** 92200133030

1) What is computer?

- A computer is an electronic device that accepts data, processes it, and produces useful information as output.
- It works based on the principle of Input → Processing → Output → Storage (IPOS cycle).
- Computers can perform both arithmetic and logical operations at very high speed.
- They are programmable, meaning a user can give instructions (software) to solve a wide variety of problems.
- A computer can store a large amount of data for long durations in secondary memory.
- It works with great accuracy and reduces human effort.
- Example: Desktop PC, Laptop, Server, Supercomputer.
- In short, a computer is a "data processing machine" used in every field like education, business, healthcare, and science.

2) Explain different types of computer with one line explanation.

I. Supercomputer – Extremely powerful machines used for scientific calculations and simulations.
II. Mainframe Computer – Large computers used in organizations for bulk data processing.
III. Mini Computer – Mid-sized machines used by small businesses or departments.
IV. Microcomputer – Personal computers like desktops and laptops for individual use.
V. Workstation – High-performance personal computers for technical/scientific work.
VI. Embedded Computer – Computers embedded inside devices like washing machines, ATMs, cars.
VII. Server – A computer that provides services/resources to other computers in a network.

3) Name various generation of computers with name of technology and example.

1. First Generation (1940–1956): Vacuum tubes, machine language. Example: ENIAC.
2. Second Generation (1956–1963): Transistors, assembly language. Example: IBM 1401.
3. Third Generation (1964–1971): Integrated Circuits (ICs), high-level languages. Example: IBM 360.
4. Fourth Generation (1971–1980s): Microprocessors, personal computers. Example: Apple II, IBM PC.
5. Fifth Generation (1980s–Present): Artificial intelligence, parallel processing. Example: Supercomputers, AI systems.

4) Explain different input and output device of computers.

Input Devices: Used to enter data into the computer.

- Keyboard (text input)
- Mouse (pointing device)
- Scanner (image/text scanning)
- Microphone (voice input)
- Webcam (video input)
- Touch screen (direct interaction)

Output Devices: Used to display processed information.

- Monitor (visual output)
- Printer (hard copy output)
- Speakers (audio output)
- Projector (large display output)
- Plotter (engineering drawings).

5) Explain following network topology with its advantage and disadvantages.

a. Bus topology

- All devices connected to a single central cable.
- Advantage: Easy to set up, cost-effective.
- Disadvantage: Failure of main cable crashes the network.

b. Ring topology

- Devices connected in a circular path.
- Advantage*:* Equal access to all devices.
- Disadvantage*:* One failure affects the entire network.

c. Star topology

- All devices connected to a central hub/switch.
- Advantage: Easy fault detection and isolation.
- Disadvantage: If hub fails, network stops.

d. Mesh topology

- Every device connected to every other device.
- Advantage: Highly reliable, no single point of failure.
- Disadvantage: Expensive, requires more cabling.

e. Hybrid Topology

- Combination of two or more topologies.
- Advantage: Flexible, scalable.
- Disadvantage: Complex and costly to maintain.

6) What is Internet, Intranet and Extranet?

- **Internet**: A global network that connects millions of computers worldwide for communication and information sharing.
- **Intranet**: A private network used within an organization for employees. Example: University intranet for students and staff.
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- **Extranet**: A controlled private network allowing access to external partners, suppliers, or customers. Example: Supplier login portal.

7) What is Information System?

- An information system (IS) is a structured setup that collects, processes, stores, and distributes information.
- It integrates people, hardware, software, databases, and networks
- Purpose: To support decision-making, coordination, and control in an organization.
- Example: Banking information system, Hospital management system.

8) Differentiate Data and Information?

- Data: Raw facts, figures, or symbols without context. Example: "90, 85, 88."
- Information: Processed and meaningful data useful for decision-making. Example: "Average score is 87."
- Data is the input; information is the output.

9) Explain hierarchy of management activity.

1. Top-level management – Strategic planning, policy-making, setting goals (CEO, Directors).
2. Middle-level management – Tactical planning, coordinating between top and lower levels (Managers, HODs).
3. Lower-level management – Operational control, daily task supervision (Supervisors, Team leaders).
   - Together, they ensure smooth functioning of an organization.

10) What is MIS?

- MIS stands for Management Information System.
- It is a computer-based system that provides information to managers for effective decision-making.
- MIS converts raw data into useful information.
- Example: Sales reports, employee performance reports.

11) Explain various component of MIS.

1. Hardware – Computers, servers, storage devices.
2. Software – Applications for data processing and reporting.
3. Database – Storage of organizational data.
4. People – Users who interact with MIS.
5. Procedures – Rules for data collection, processing, and usage.

12) What is the role of MIS?

1. Supports planning, controlling, and decision-making.
2. Provides timely and accurate reports.
3. Improves efficiency by automating data handling.
4. Facilitates smooth communication in organization.

13) Explain main objectives of MIS.

- Provide relevant and timely information.
- Improve decision-making.
- Enhance productivity and efficiency.
- Support strategic planning.
- Maintain data accuracy and security.

14) Explain main characteristics of MIS.

- Timely, accurate, and relevant.
- User-friendly interface.
- Flexible and adaptable to change.
- Provides both routine and special reports.
- Integrates all departments of organization.

15) What is impact of MIS on any organization?

- Improves decision-making quality.
- Increases operational efficiency.
- Provides competitive advantage.
- Enhances communication and transparency.
- Reduces redundancy and errors.

16) What are the advantages and disadvantages of MIS?

Advantages:
- Quick access to data
- Better decision-making
- Cost savings
- Improved productivity
- Enhanced communication

Disadvantages:
- High cost of implementation
- Requires skilled manpower
- Risk of system failure
- Security concerns

17) Explain TPS with its component and its types.

- **Transaction Processing System (TPS):** Manages daily routine transactions of an organization.
- **Components:** Input (data capture), Processing (validation), Output (receipts, reports), Database (storage).

- **Types:**
  - Batch Processing (transactions processed in groups, e.g., payroll).
  - Real-time Processing (immediate update, e.g., ATM withdrawals).

18) Explain DSS with its various model and example.

- **Decision Support System (DSS):** Helps managers in semi-structured decision-making.
- **Models**:
  1. **Data-driven DSS** – Focuses on database queries (sales analysis).
  2. **Model-driven DSS** – Uses mathematical/analytical models (forecasting).
  3. **Knowledge-driven DSS** – Uses rules and AI (expert systems).
  4. **Communication-driven DSS** – Supports group decision-making (video conferencing).

19) What is Executive Information System?

- A system designed for top executives to monitor performance and analyze trends.
- Provides easy-to-read dashboards and summaries.
- Helps in long-term strategic planning.
- Example: CEO dashboard showing revenue growth, market share.

20) Differentiate DSS and ESS.

- DSS: For middle-level managers, semi-structured decisions, detailed reports.
- ESS (EIS): For top-level executives, strategic unstructured decisions, graphical summaries.

21) What is Knowledge Management System with types of Knowledge?

- KMS captures, organizes, and shares organizational knowledge.
- **Types**:
  - **Explicit Knowledge**: Documented (manuals, reports).
  - **Tacit Knowledge**: Personal experience and skills.

22) Explain benefits of KMS.

- Encourages innovation.
- Improves decision-making.
- Reduces redundancy.
- Enhances employee collaboration.
- Increases organizational learning.

23) What is Functional Information system?

- Specialized IS for specific functions of business like marketing, finance, HR, or manufacturing.
- Supports departmental operations and decision-making.

24) Explain various component for given Information System.

a. **Marketing Information System :-** Sales reports, customer trends, market research.
b. **Financial Information System :-** Budgeting, accounts, financial forecasting.
c. **Human Resource Information System :-** Payroll, recruitment, performance tracking.
d. **Manufacturing Information System :-** Production scheduling, inventory control, supply chain.

25) What is meaning of Ethics?

- Ethics is a set of moral principles that govern human behavior.
- It guides what is right or wrong in personal and professional life.

26) What is Business Ethics?

- Application of ethical principles in business environment.
- Ensures fairness, honesty, and responsibility in all dealings.

27) What is the need and importance of Business Ethics at workplace?

- Builds trust with customers.
- Improves company reputation.
- Reduces legal issues.
- Promotes healthy workplace culture.
- Ensures long-term sustainability.

28) Discuss various Ethical issues in Business.

- Bribery and corruption.
- False advertising.
- Misuse of company resources.
- Exploitation of employees.
- Environmental pollution.

29) Explain importance of Ethical responsibility of business professional.

- Maintain honesty and integrity.
- Respect confidentiality.
- Follow laws and regulations.
- Avoid conflicts of interest.
- Contribute positively to society.

30) What is Cyber Crime?

- Cybercrime refers to criminal activities carried out using computers or the internet.
- Examples: Hacking, phishing, identity theft, cyberbullying, ransomware.
- It can target individuals, organizations, or governments.
- Cybercrime is a growing threat that requires strong cybersecurity measures.